StudySapiens Board MCQs

Chapter 7 — Structural and Computational Biology

Board-style MCQs · Questions first, answer key with brief reasons at the end · PECTAA 2025–26

Choose the best option. Each question carries one mark. The answer key with brief reasons is given at the end of this chapter.

1. Structural biology deals with the study of:

(a) Only DNA sequences
(b) Only cell division
(c) Three-dimensional structures of macromolecules
(d) Only ecology

2. Structural biology is especially useful in:

(a) Soil formation only
(b) Medical research and drug design
(c) Weather prediction
(d) Taxonomy only

3. The 3D structure of HIV-1 reverse transcriptase helped identify its:

(a) Polymerase and RNase H domains
(b) Capsid and envelope
(c) Cell wall and membrane
(d) Ribosomes and lysosomes

4. A drug target is usually a:

(a) Water molecule
(b) Carbohydrate only
(c) Protein
(d) Mineral ion only

5. The SARS-CoV-2 spike protein is important because it:

(a) Produces ATP
(b) Fixes nitrogen
(c) Forms the cell wall
(d) Helps the virus enter human cells

6. Structural biology can help block viral entry by designing drugs that:

(a) Destroy host DNA
(b) Increase viral replication
(c) Inhibit virus-receptor interaction
(d) Form viral capsids

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7. Protein misfolding is associated with diseases such as:

(a) Malaria only
(b) Influenza only
(c) Alzheimer’s and Parkinson’s
(d) Tuberculosis only

8. X-ray crystallography was developed by:

(a) William Henry Bragg and William Lawrence Bragg
(b) Mendel and Morgan
(c) Singer and Nicolson
(d) Watson and Crick

9. In X-ray crystallography, the protein must first be:

(a) Crystallized
(b) Hydrolysed
(c) Denatured
(d) Fermented

10. The diffraction pattern produced by X-rays is used to create a/an:

(a) Genetic map
(b) Ecological pyramid
(c) Action spectrum
(d) Electron density map

11. The final 3D protein structure in X-ray crystallography is obtained by:

(a) Light microscopy
(b) Centrifugation only
(c) Computational analysis of diffraction data
(d) Gram staining

12. Computational biology integrates biology with:

(a) Only physics
(b) Computer science, mathematics and statistics
(c) Only geology
(d) Only chemistry

13. Genomics is the study of:

(a) Genomes
(b) Cells only
(c) Proteins only
(d) Lipids only

14. Proteomics is the large-scale study of:

(a) Proteins
(b) RNA only
(c) Carbohydrates only
(d) DNA only

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15. Bioinformatics is mainly concerned with:

(a) Microscope construction
(b) Animal dissection
(c) Storage, retrieval and analysis of biological data
(d) Cell staining only

16. Computational biology helps drug discovery by:

(a) Predicting drug-target interactions
(b) Forming crystals only
(c) Replacing all experiments
(d) Producing antibiotics directly

17. GenBank is a database of:

(a) Carbohydrates only
(b) Nucleotide sequences
(c) Protein structures only
(d) Lipids only

18. The Protein Data Bank (PDB) stores:

(a) 3D structures of proteins and nucleic acids
(b) Only ecological data
(c) Only metabolic pathways
(d) Only DNA sequences

19. Ensembl is a:

(a) Enzyme class
(b) Protein stain
(c) Genome browser
(d) Microscope

20. BLAST is used to:

(a) Crystallize proteins
(b) Measure blood pressure
(c) Stain cells
(d) Compare biological sequences

21. FASTA is a tool for:

(a) Protein digestion
(b) Cell culture
(c) X-ray diffraction
(d) Sequence alignment

22. Sequence homology refers to similarity due to:

(a) Same length only
(b) Random chance only
(c) Same colour only
(d) Shared ancestry

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23. Orthologs are homologous sequences found in:

(a) The same species after gene duplication
(b) Viruses only
(c) Different species after speciation
(d) Plasmids only

24. Paralogs arise by:

(a) Speciation only
(b) Protein denaturation
(c) Gene duplication within a species
(d) Viral infection

25. Orthologs often:

(a) Retain the same function across species
(b) Become lipids
(c) Always lose function
(d) Form viruses

26. Paralogs may:

(a) Evolve new functions
(b) Always cause disease
(c) Never change function
(d) Only form RNA

27. If an unknown gene is homologous to a gene of known function, it likely:

(a) Has a similar function
(b) Is viral
(c) Has no function
(d) Is a lipid

28. Structural homology refers to similarity in:

(a) Three-dimensional structure
(b) Only colour
(c) Only size
(d) Only nucleotide sequence

29. Proteins with similar structures often:

(a) Perform similar functions
(b) Are always enzymes
(c) Are always lipids
(d) Always have identical sequences

30. Structural homology is useful in:

(a) Soil classification
(b) Grammar checking
(c) Drug design
(d) Weather forecasting

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31. Model organisms are useful in disease research because:

(a) They are always human
(b) They lack proteins
(c) They have no genes
(d) Homologous genes can reveal human disease mechanisms

32. The main advantage of computational biology is its ability to:

(a) Produce cells
(b) Form crystals
(c) Handle large datasets and find patterns
(d) Replace microscopes completely

33. A domain in a protein is:

(a) A sugar chain
(b) A nucleotide only
(c) A lipid layer
(d) A distinct structural unit with an independent function

34. The active site of an enzyme can be identified using:

(a) Only fermentation
(b) Structural biology
(c) Only centrifugation
(d) Only Gram staining

35. Which database would you use to find a protein's 3D structure?

(a) BLAST only
(b) PDB
(c) Ensembl only
(d) GenBank only

36. Which tool would you use to find similar DNA sequences?

(a) Sphygmomanometer
(b) BLAST
(c) X-ray crystallography
(d) Electrophoresis only

37. Personalized medicine can be supported by computational biology through:

(a) Growing crystals only
(b) Measuring transpiration
(c) Analysis of genetic variations associated with disease
(d) Staining tissues only

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38. Evolutionary relationships can be reconstructed by:

(a) Measuring cell size only
(b) Comparing genetic information across species
(c) Counting chromosomes only
(d) Observing colour only

39. The Nobel Prize for X-ray crystallography was awarded in:

(a) 1915
(b) 1972
(c) 1990
(d) 1953

40. A major application of structural biology is:

(a) Forming annual rings
(b) Identifying host-pathogen interactions
(c) Fixing nitrogen
(d) Producing ATP

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ANSWER KEY WITH BRIEF REASONS

1. (c) Three-dimensional structures of macromolecules — It studies proteins/nucleic acids at atomic level.

2. (b) Medical research and drug design — 3D structures reveal drug targets.

3. (a) Polymerase and RNase H domains — Domain knowledge aids antiviral design.

4. (c) Protein — Drugs often bind specific proteins.

5. (d) Helps the virus enter human cells — Spike mediates host-cell entry.

6. (c) Inhibit virus-receptor interaction — Blocking binding prevents infection.

7. (c) Alzheimer's and Parkinson's — Misfolded proteins cause several disorders.

8. (a) William Henry Bragg and William Lawrence Bragg — The Braggs developed the technique.

9. (a) Crystallized — Crystals give ordered diffraction.

10. (d) Electron density map — Density maps reveal atom positions.

11. (c) Computational analysis of diffraction data — Mathematical reconstruction builds the model.

12. (b) Computer science, mathematics and statistics — It uses computational tools for biological data.

13. (a) Genomes — Genomics analyses complete DNA sets.

14. (a) Proteins — Proteomics studies the proteome.

15. (c) Storage, retrieval and analysis of biological data — It manages sequence/structure data.

16. (a) Predicting drug-target interactions — It simulates and predicts binding.

17. (b) Nucleotide sequences — GenBank stores DNA/RNA sequences.

18. (a) 3D structures of proteins and nucleic acids — PDB is a structural database.

19. (c) Genome browser — Ensembl visualizes genomic data.

20. (d) Compare biological sequences — BLAST finds sequence similarity.

21. (d) Sequence alignment — FASTA compares query sequences to databases.

22. (d) Shared ancestry — Homologous sequences evolve from a common ancestor.

23. (c) Different species after speciation — Orthologs arise by speciation.

24. (c) Gene duplication within a species — Paralogs are duplication-derived.

25. (a) Retain the same function across species — Conserved function is common.

26. (a) Evolve new functions — Duplication allows functional divergence.

27. (a) Has a similar function — Homology suggests functional similarity.

28. (a) Three-dimensional structure — Structure can be conserved despite sequence divergence.

29. (a) Perform similar functions — Structure determines function.

30. (c) Drug design — Drugs target protein structures.

31. (d) Homologous genes can reveal human disease mechanisms — Conserved genes allow inference.

32. (c) Handle large datasets and find patterns — It analyses big biological data.

33. (d) A distinct structural unit with an independent function — Domains are functional/structural modules.

34. (b) Structural biology — 3D structure reveals active sites.

35. (b) PDB — PDB contains structural data.

36. (b) BLAST — BLAST compares sequences.

37. (c) Analysis of genetic variations associated with disease — Genomic data guide tailored therapy.

38. (b) Comparing genetic information across species — Sequence comparison reveals ancestry.

39. (a) 1915 — The Braggs received the 1915 Physics Nobel.

40. (b) Identifying host-pathogen interactions — It shows how pathogens bind host cells.

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